



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 4 • STANDARD 6.AT.2

# Rates and Unit Rates

Flagship

Lesson 4-1-flagship · Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## ARCADE BUILDER MISSION

### Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges \$3 for 5 games, Booth B charges \$5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can find a unit rate to compare prices and decide the better buy.

**Language Objective:** I can explain my choice using the words rate, unit rate, per, and better buy.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rate

Unit Rate

Per

Ratio

Better Buy

Divide



Tap any word to see what it means and a picture.

1

A ratio that compares two quantities with different units is a

.

2

A rate that tells the cost or amount for one unit is a  .

3

The word that means 'for each one' is  .

4

A comparison of two quantities is a  .

5

The option with the lower unit rate, costing less per item, is the

.

6

To find a unit rate, I  the price by the number of units.

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### EXPLAIN

**How did you find the cost per game for each booth, and which booth wins?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

☐ **Rate** — A ratio comparing two amounts with different units, like miles per hour.  
*Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.*

☐ **Unit Rate** — A rate for just 1 of something, like cost for 1 item.  
*Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.*

☐ **Per** — For each one. Example: 5 dollars per book.  
*Por — Por cada uno. Ejemplo: 5 dólares por libro.*

☐ **Better Buy** — The choice that costs less for each item.  
*Mejor compra — La opción que cuesta menos por cada artículo.*

☐ **Divide** — Splitting a total into equal groups.  
*Dividir — Repartir un total en grupos iguales.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**Booth A costs \_\_\_\_ per game because 3 divided by 5 equals \_\_\_\_.**

*El puesto A cuesta \_\_\_\_ por juego porque 3 entre 5 es \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** You can't compare fairly until you find the cost per 1 game (the unit rate) for each booth.

**Evidence:** Students recognize the booths offer different numbers of games, so a fair comparison needs the cost per 1 game (unit rate).

**Reasoning:** This evidence connects the definition of rate to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Booth A costs \_\_\_\_ per game because 3 divided by 5 equals \_\_\_\_.**

*El puesto A cuesta \_\_\_\_ por juego porque 3 entre 5 es \_\_\_\_.*

- **The better buy is Booth \_\_\_\_ because its \_\_\_\_ is lower.**

*La mejor compra es el puesto \_\_\_\_ porque su \_\_\_\_ es más baja.*

#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Rate
2. **Notes 2:** Unit Rate
3. **Notes 3:** Per
4. **Notes 4:** Ratio
5. **Notes 5:** Better Buy
6. **Notes 6:** Divide
7. **Watch for this mistake:** *A common mistake in Rates and Unit Rates is dividing the numbers in the wrong order — for example, finding  $\text{games} \div \text{cost}$  instead of  $\text{cost} \div \text{games}$ . This flips the rate (games per dollar instead of dollars per game) and gives a number that looks reasonable but answers the wrong question. Another common mistake is comparing the total prices of two deals instead of their unit rates — a bigger total cost does not always mean a worse deal. Before you answer, ask: 'What is 1 unit of, and did I divide the cost by that unit?'*
8. **Try It 1:** A. Pixel Racer (50 points per minute) — *Pixel Racer:  $250 \div 5 = 50$  points per minute. Turbo Drift:  $180 \div 4 = 45$  points per minute.  $50 > 45$ , so Pixel Racer is faster.*
9. **Try It 2:** A. \$5.00 for 10 plays —  $\$5.00 \div 10 = \$0.50$  per play;  $\$6.00 \div 8 = \$0.75$  per play;  $\$9.00 \div 12 = \$0.75$  per play;  $\$7.00 \div 10 = \$0.70$  per play. The lowest unit rate, \$0.50 per play, is the best value.